• Apply the research ethics during the entire research work.	**6. Ethical Issues in Research (4 hrs)** 6.1. Right of People directly involved 6.2. Responsibilities of an Ethical Researcher 6.3. Data Privacy and Security 6.4. Plagiarism and Intellectual Property 6.5. Algorithmic Bias and Fairness 6.6. Environmental Responsibility 6.7. Conflict of Interest 6.8. Copyright and Patent 6.9. ACM Code of Ethics and IEEE Code of Ethics

1. **As an early-stage researcher, how do you write and present your research? Explain.**

For an early-stage researcher, writing and presenting research is the critical final stage—the **Presentation**—where work is communicated to the academic community for scrutiny and evaluation. This process involves developing specific skills in academic writing and speaking, following established structures, and adhering to rigorous ethical standards.

Strategies for Writing Your Research

Effective scholarly writing is an iterative process that serves as a tool for thinking and clarifying ideas.

- **Writing as Thinking:** You should **write ideas down early** in the research process rather than just talking about them. Drafting a one-to-two-page overview helps visualize the final product and allows advisers to provide better feedback.
- **Establishing a Habit:** Develop a **regular and continuous habit of writing** each day. Avoid becoming a "weekend writer," as start-and-stop processes can disrupt the flow of work.
- **Drafting and Polishing:** Do not try to polish the first draft. Follow a three-stage model: first develop an **outline or visual map**, then write a rough draft and move paragraphs to sort ideas, and only in the final stage **edit and polish** each sentence.
- **Enhancing Readability:**
    - **Coherence:** Use the **"hook-and-eye" technique** to ensure every sentence is a logical sequel to the one before it, connecting key ideas across paragraphs.
    - **Voice and Tense:** Use **active voice** as much as possible. Use the **past tense** to review literature or report results, and the **present tense** to discuss results and conclusions.
    - **Trimming "Fat":** Systematically edit to remove unnecessary words and modifiers that do not add meaning. Use **consistent terms** for variables rather than synonyms to avoid confusing the reader.

Structure of the Research Account

The structure of your writing depends on the research design and the target format (e.g., thesis, journal, or conference paper).

- **Conventional Structure:** Most research includes a **Title, Abstract, Introduction, Literature Review, Methodology, Results, Discussion, and Conclusion**.
- **Quantitative vs. Qualitative:** Quantitative reports have a set structure with separate results and discussion sections. Qualitative reports are more flexible and often **integrate data (like quotes) directly with interpretation**.
- **Referencing:** You must use a consistent style manual, such as **IEEE** (common in computer science) or **APA**. In IEEE style, citations are **numbered in the order of appearance** and listed numerically at the end.

Presenting Research Outcomes

Presentations can take written, visual, or oral forms depending on the audience and venue.

- **Written Presentations:**
    - **Journal Papers:** These provide comprehensive, finalized research with in-depth analysis and are considered authoritative.
    - **Conference Papers:** These typically present **preliminary or ongoing research** and are shorter (4-10 pages).
    - **Technical Reports:** Detailed documents communicating processes and findings for specialized or industry audiences.
- **Oral Presentations:** Usually 20 minutes long with 10 minutes for Q&A. You should **avoid reading your paper verbatim**; instead, use 10-14 slides to highlight key arguments and visual data like screenshots or code segments.
- **Visual Presentations:**
    - **Posters:** Summarize research using eye-catching visuals and minimal text, legible from a distance.
    - **Infographics:** Ideal for simplifying complex data into trends and comparisons for non-expert audiences.

Ethical Considerations

Integrity is the foundation of research presentation. Researchers must **avoid plagiarism** by never cutting and pasting from other sources or re-using their own previously marked work without citation (self-plagiarism). You must also **disclose conflicts of interest**, acknowledge all contributors through proper authorship, and provide accurate accounts of your data without falsification.

2. **What do you mean by data privacy and copyright?How can researchers maintain objectivity when their research is funded by private corporations with vested interests?**

Data privacy in research refers to the ethical and legal obligation to handle participants' personal information with care, respecting their right to confidentiality throughout the research process. It involves complying with regulations such as **GDPR and HIPAA**, ensuring **informed consent**, and using techniques like **anonymization** to remove or disguise identifiable information. Advanced computational methods, such as **Federated Learning**, are increasingly used to protect privacy by training models on local devices so that sensitive data never leaves the client. **Security** measures, including **encryption**, password protection, and secure servers, are essential to prevent unauthorized access or data breaches.

Copyright is an intangible right granted to the author or originator of original literary or artistic productions. Under the **Berne Convention**, this protection exists automatically upon the creation of a work without the need for formal registration. In the context of research, copyright applies to **images, software, theses, and publications**. Key characteristics include:

- **Duration:** Statutory protection generally lasts for the life of the author plus **70 years**.
- **Ownership:** Employers typically own "works for hire" created within the scope of employment unless otherwise agreed.
- **Usage:** Researchers must obtain **written permission** to reproduce copyrighted materials and must always provide proper attribution to avoid plagiarism.
- **Limitations:** Facts themselves are not copyrightable.

Maintaining Objectivity in Funded Research

When research is funded by private corporations with vested interests, maintaining **objectivity**—defined as letting facts speak for themselves and avoiding bias—is a critical ethical responsibility. Researchers can employ several strategies to safeguard the integrity of their findings:

3. **Full Disclosure of Funding:** Researchers must explicitly specify the **sponsorship** of their study to establish trust and transparency. This includes declaring any **conflicting financial interests** or personal relationships that could appear to influence the work.
4. **Managing Conflicts of Interest (COI):** A COI occurs when financial or personal factors impair professional judgment. Researchers and institutions should follow established policies to **identify, minimize, or eliminate** these conflicts.
5. **Adherence to Professional Codes:** Following ethical guidelines from organizations like **IEEE** or **ACM** provides a framework for professional conduct. The IEEE Code of Ethics specifically mandates avoiding real or perceived conflicts of interest and disclosing them to affected parties when they exist.
6. **Honest and Accurate Reporting:** Researchers have a responsibility to record and report data **fully and accurately**, avoiding the temptation to manipulate or omit results that do not support a sponsor's interests.

7. **Ethical Oversight:** Submitting research plans to an **Institutional Review Board (IRB)** or ethics committee ensures external scrutiny of the project's design and funding arrangements.
8. **Reflexivity:** Researchers should reflect on and explicitly identify their own **biases and values** at the outset of a study, acknowledging how their background may shape their interpretations.

9. **Describe the importance of plagiarism, conflict of interest, patent, and codes of ethics in research.**

In research, maintaining integrity and upholding standards is facilitated by clear definitions and enforcement of ethical boundaries. The following concepts—plagiarism, conflict of interest, patents, and codes of ethics—are vital to ensuring that research remains a credible and beneficial endeavor for both the academic community and society.

1. Plagiarism: Safeguarding Originality and Honesty

Plagiarism is defined as the appropriation of another person's ideas, processes, results, or words without giving appropriate credit. It is essentially the **theft of intellectual property** and is considered a form of scientific misconduct.

- **Importance:** Avoiding plagiarism is a core ethical responsibility because it **undermines trust, originality, and intellectual growth**.
- **Consequences:** Engaging in plagiarism leads to severe academic and professional penalties, including failing grades, expulsion, job loss, and public humiliation. It also stifles innovation by devaluing original ideas.
- **Self-Plagiarism:** Researchers must also avoid "re-using own work which was previously marked" or published without proper disclosure, as this is viewed as **self-plagiarism** and misleads the audience about the novelty of the findings.

2. Conflict of Interest: Ensuring Objectivity and Trust

A **conflict of interest (COI)** occurs when personal, financial, or professional interests interfere, or appear to interfere, with the integrity and impartiality of research.

- **Importance:** Managing conflicts is essential because unmanaged COIs cast doubt on **research impartiality** and decision-making.
- **Impact on Credibility:** Conflicts can erode public trust and undermine the credibility of scientific findings. The sources emphasize that the **perception of dishonesty** can be just as damaging as actual misconduct.
- **Mitigation:** Researchers and institutions have a mandatory obligation to **disclose real or perceived conflicts** to maintain transparency and accountability.

3. Patents: Incentivizing and Protecting Innovation

A **patent** is a legal monopoly granted by a government that allows an inventor exclusive rights to make, use, and sell an invention for a specific period (typically 20 years).

- **Importance:** Patents are critical for protecting **industrial property** and providing the economic incentive needed for research and development (R&D).
- **Encouraging Growth:** By ensuring that creators can profit from their work and preventing "free loaders," patents facilitate the **transfer of technology** and cultural development.
- **Criteria:** To be granted a patent, an invention must be **novel, non-obvious, and useful**.

4. Codes of Ethics: Providing a Framework for Conduct

Codes of ethics are sets of rules and principles drafted by professional associations—such as **ACM** and **IEEE**—to govern scholarly research and professional behavior.

- **Importance:** These codes ensure **integrity, quality, and accountability** by providing a standardized framework for ethical decision-making.
- **Guiding Research Design:** They assist researchers in navigating complex issues such as protecting the **privacy of others**, handling sensitive information, and working with vulnerable populations.
- **Professional Responsibility:** Adhering to these codes mandates that researchers hold the **safety, health, and welfare of the public** paramount, treat all persons fairly, and seek honest criticism of their technical work.

In summary, while plagiarism and conflict of interest policies act as **shields against bias and dishonesty**, patents and codes of ethics serve as **incentives for innovation** and blueprints for **responsible professional behavior**.

10. Ethical Issues in Research

Ethical issues in research involve a set of values, standards, and principles used to distinguish between acceptable and unacceptable conduct at every stage of the scientific process. Upholding research ethics is essential to ensure **integrity, quality, and accountability**, as well as to protect the rights and well-being of all participants involved.

Core Ethical Values and Responsibilities

The foundation of research ethics rests on shared values, including **trust, honesty, accuracy, and objectivity**. Researchers have a responsibility to:

- **Act with Integrity:** Report data fully and accurately, avoiding the temptation to manipulate, falsify, or omit results that do not support a specific case.
- **Avoid Unnecessary Intrusion:** Ensure the research is not redundant and minimize intrusive questions by using techniques like age ranges instead of exact dates.
- **Adhere to Professional Codes:** Follow guidelines from professional bodies such as the **ACM** or **IEEE**, which mandate public safety, fairness, and the avoidance of bribery or harassment.
- **Be Ethical Reviewers:** Provide objective, constructive feedback while maintaining the confidentiality of manuscripts under review.

Protecting the Rights of Participants

All research participants have fundamental human rights that must be respected regardless of the research strategy.

- **Informed Consent:** Participants must be fully informed about the study's purpose, the researchers, sponsoring organizations, and potential risks before signing an **informed consent form**. Deception regarding the study's true intent is considered unethical.
- **Voluntary Participation and Withdrawal:** Individuals have the right to decline participation without coercion and to withdraw from a study at any stage without penalty.
- **Anonymity and Confidentiality:** Researchers must protect participant identities—often through **pseudonyms**—and ensure that data is stored securely and discarded after a reasonable period (typically 5–10 years).

Scientific Misconduct and Intellectual Property

Maintaining the credibility of the academic community requires strict adherence to rules regarding intellectual property.

- **Plagiarism:** This is defined as the theft of ideas, words, or results without proper credit and is a form of **scientific misconduct**. It includes "self-plagiarism," where a researcher reuses their own previously submitted work without disclosure.
- **Conflict of Interest (COI):** A COI occurs when personal or financial factors impair a researcher’s professional judgment. Researchers are mandated to disclose any sponsorship or vested interests to maintain transparency.
- **Authorship:** It is unethical to engage in "gift authorship" for those who did not contribute or "ghost authorship" that omits junior staff who made significant contributions.

Emerging Ethical Issues in IT Research

Information technology research presents unique ethical challenges related to digital data and automated systems.

- **Algorithmic Bias and Fairness:** Algorithms can produce discriminatory outcomes if they are trained on biased data or developed by non-diverse teams. Researchers must use representative datasets and implement bias-detection tools to ensure fairness.
- **Data Privacy and Security:** With the rise of internet-based data collection, researchers must comply with laws like **GDPR** and use encryption and anonymization to prevent data breaches.
- **Environmental Responsibility:** This involves an ethical obligation to minimize the negative impact of research activities on ecosystems, such as regulating energy consumption in large-scale computational tasks.

Ethical Oversight

Before beginning any study, researchers must obtain **ethical clearance** from an **Institutional Review Board (IRB)** or Research Ethics Committee. This board assesses potential risks—physical, psychological, social, or legal—to ensure the study complies with human rights protections. Special care is required for **vulnerable populations**, such as minors, prisoners, or those with neurological impairments.

11. **Research Ethics**
12. **As an ethical researcher, what are your key responsibilities? Explain the role of intellectual property rights (IPR) in scientific research.**

As an ethical researcher, your key responsibilities span the protection of participant rights, the maintenance of professional integrity, and the advancement of knowledge while adhering to legal and institutional standards.

Key Responsibilities of an Ethical Researcher

- **Respect for Participant Rights:** You must ensure that participation is voluntary and that individuals have the right to withdraw at any stage. This includes obtaining **informed consent** by fully disclosing the research purpose, risks, and data usage. Additionally, you are responsible for protecting the **anonymity and confidentiality** of participants through the use of pseudonyms and secure data handling.
- **Behavior with Integrity:** You must record and report data accurately and fully, avoiding any **manipulation, falsification, or fabrication** of findings. This also involves being transparent about any **sponsors** or potential **conflicts of interest** that might appear to influence your work.
- **Avoidance of Unnecessary Intrusion:** Ethical research should not be redundant; you must ensure the information is not already available and minimize intrusive questions by using techniques like age ranges instead of exact dates.
- **Adherence to Professional Codes:** You are expected to follow the standards established by professional bodies such as the **ACM** or **IEEE**. For example, the IEEE

Code of Ethics mandates holding the safety, health, and welfare of the public paramount and seeking honest criticism of technical work.
- **Commitment to the Academic Community:** This includes avoiding **plagiarism**, providing proper attribution to others' work, and being an **ethical reviewer** by completing timely, objective, and confidential reviews.

The Role of Intellectual Property Rights (IPR) in Scientific Research

Intellectual Property Rights represent the legal and ethical framework for protecting creations of the mind, which are critical for incentivizing and disseminating scientific innovation.

- **Protecting Originality and Encouraging Innovation:** IPR, such as **patents** and **copyright**, provides the economic incentive needed for research and development (R&D) by ensuring creators can profit from their work and preventing "free loaders". Patents grant a legal monopoly (typically for 20 years) for novel, non-obvious, and useful inventions.
- **Safeguarding Intellectual Assets:** Copyright automatically protects original literary and artistic productions, including **software, images, theses, and publications**, from the moment of creation. In research, this ensures that your written accounts and technical artifacts are legally recognized as your intellectual property.
- **Preventing Scientific Misconduct:** Respecting IPR is essential to avoid **plagiarism**, which is defined as the theft of intellectual property (ideas, words, or results) without giving appropriate credit. Proper citation and the use of quotation marks for direct quotes are necessary to honor property rights and maintain academic credibility.
- **Facilitating Technology Transfer:** By clarifying ownership, IPR allows for the legal sale, lease, or assignment of research outcomes, thereby facilitating the **transfer of technology** and cultural development across society.